MATH441/MATH541
Applied Mathematics II: Complex Variables
Lecture slides 7: Complex integrals: Line integrals

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Lecture outline

1. Rectifiable curves
2. Line integrals
3. Properties of integrals
4. Smooth curves

Rectifiable curves. Part 1.

Let L be a continuous curve with equation

$$z = \lambda(t) = \mu(t) + i\nu(t) \ \ (a \leq t \leq b), \tag{1}$$

and suppose we divide the interval $[a, b]$ into n subintervals

$$[t_{k-1}, t_k] \ \ (k = 1, \ldots, n) \tag{2}$$

by introducing $n - 1$ intermediate points $t_1, \ldots, t_{n-1}$ satisfying the inequalities

$$a = t_0 < t_1 < \cdots < t_{n-1} < t_n = b.$$

The set

$$\mathcal{P} = \{t_0, t_1, \ldots, t_n\} \tag{3}$$

is called a *partition* of the interval $[a, b]$, and the largest of the numbers

$$t_1 - t_0, \ t_2 - t_1, \ \ldots, \ t_n - t_{n-1},$$

i.e. the maximum length of the subintervals (2) is called the *norm* of the partition $\mathcal{P}$, denoted by $|\mathcal{P}|$.

Rectifiable curves. Part 2.

Corresponding to the partition (3), the curve L is divided into n arcs

$$\sigma_k = \widehat{z_{k-1}z_k} \quad (k = 1, \ldots, n), \tag{4}$$

where $\widehat{z_{k-1}z_k}$ means a curve joining z_k and z_{k+1} and

$$z_k = \lambda(t_k) \quad (k = 0, 1, \ldots, n) \tag{5}$$

the final point of each arc (except the last one) coincides with the initial point of the next arc. Joining each of the points $z_0, z_1, \ldots, z_n$ to the next by a straight line segments we obtain a polygonal curve Λ *inscribed* in L. The segments of Λ are the chords joining the end points of the arcs σ_k, and the length of Λ is obviously

$$\sum_{k=1}^{n} |z_k - z_{k-1}| \tag{6}$$

Rectifiable curves. Part 3.

The curve L is said to be *rectifiable* if

$$\sup_{\mathcal{P}} \sum_{k=1}^{n} |z_k - z_{k-1}| = l < \infty,$$

where the least upper bound (denoted by the symbol sup) is taken over all possible partitions (3). The nonnegative real number l is called the *length* of the curve L. However, if the sums (6) become arbitrary large for suitably chosen partitions, the curve is said to be *nonrectifiable*. In this case, L is considered to have no length at all (or, if preferred, infinite length).

Line integrals. Part 1.

Now let L be a rectifiable curve with equation (1), and let $P(x, y)$ and $Q(x, y)$ be two real functions, defined and continuous at every point $z = x + iy$ of L. Given an arbitrary partition $\mathcal{P} = \{t_0, t_1, \ldots, t_n\}$ of the interval $[a, b]$, let σ_k and z_k be the same as in (4) and (5), and let

$$\zeta_k = \xi_k + i\eta_k = \lambda(\tau_k) \ \ (t_{k-1} \leq \tau_k \leq t_k)$$

be an arbitrary point of σ_k $(k = 1, \ldots, n)$. Then it can be shown similarly to elementary calculus that the sum

$$\sum_{k=1}^{n} [P(\xi_k, \eta_n)(x_k - x_{k-1}) + Q(\xi_k, \eta_k)(y_k - y_{k-1})] \tag{7}$$

approaches a unique limit, denoted by

$$\int_L \{P(x, y)dx + Q(x, y)dy\}, \tag{8}$$

as the norm of the partition $\mathcal{P}$ approaches zero. An expression of the form (8) is known as a line integral along the curve L.

Line integrals. Part 2.

Similarly, let

$$f(z) = u(x, y) + iv(x, y)$$

be a complex function, defined and continuous on L, and consider the sum

$$S = \sum_{k=1}^{n} f(\zeta_k)(z_k - z_{k-1}).$$

Introducing the additional notation

$$u_k = u(\xi_k, \eta_k),\ v_k = v(\xi_k, \eta_k),\ \Delta x_k = x_k - x_{k-1},\ \Delta y_k = y_k - y_{k-1},$$

Line integrals. Part 3.

we have

$$f(\zeta_k) = u_k + iv_k, \quad z_k - z_{k-1} = \Delta x_k + i\Delta y_k,$$

and hence

$$\begin{aligned} S &= \sum_{k=1}^{n} f(\zeta_k)(z_k - z_{k-1}) \\ &= \sum_{k=1}^{n} (u_k + iv_k)(\Delta x_k + i\Delta y) \\ &= \sum_{k=1}^{n} (u_k \Delta x_k - v_k \Delta y_k) + i \sum_{k=1}^{n} (v_k \Delta x_k + u_k \Delta y_k). \end{aligned} \tag{9}$$

Line integrals. Part 4.

Thus we see that the real and imaginary parts of S are both sums of the form (7), corresponding to the partition $\mathcal{P}$, where in the first case the role of the real functions P and Q is played by the functions u and $-v$, and in the second case by the functions v and u. By the (contour) integral of $f(z)$ along the curve (or contour) L, denoted by

$$\int_L f(z)dz, \quad or \quad \oint_L f(z)dz, \tag{10}$$

we mean the quantity

$$\lim_{|\mathcal{P}|\to 0} \sum_{k=1}^{n} f(\zeta_k)(z_k - z_{k-1}), \tag{11}$$

i.e., the limit of the sum S as the norm of the partition $\mathcal{P}$ approaches zero.

Line integrals. Part 5.

Since the real functions u and v are continuous on L (being the real and imaginary parts of the continuous function $f(z)$), and since L is rectifiable, it follows from (9) and the definition of the corresponding line integrals that

$$\int_L f(z)dz = \int_L \{u(x,y)dx - v(x,y)dy\} + i\int_L \{v(x,y)dx + u(x,y)dy\}. \quad (12)$$

In other words, evaluating the integral of a complex function reduces to evaluating two line integrals of real functions.

Example 1.

Example

If $f(z) = 1$, (10) becomes

$$\int_L dz = \lim_{|\mathcal{P}|\to 0} \sum_{k=1}^{n} (z_k - z_{k-1}) = \lim_{|\mathcal{P}|\to 0} (z_n - z_0) = Z - z_0,$$

where z_0 is the initial and Z the final point of the curve L. In particular, if L is closed, $Z = z_0$ and

$$\oint_L dz = 0$$

Example 2.

Example

If $f(z) = z$, we first choose ζ_k to be initial point z_{k-1} of the arc $\sigma_k = \widehat{z_{k-1}z_k}$, and then the final point z_k of σ_k. This gives

$$\begin{aligned}
\int_L z dz &= \lim_{|\mathcal{P}|\to 0} \sum_{k=1}^{n} z_k(z_k - z_{k-1}) \\
&= \frac{1}{2} \lim_{|\mathcal{P}|\to 0} \sum_{k=1}^{n} (z_k + z_{k-1})(z_k - z_{k-1}) \\
&= \frac{1}{2} \lim_{|\mathcal{P}|\to 0} \sum_{k=1}^{n} (z_k^2 - z_{k-1}^2) \\
&= \frac{1}{2} \lim_{|\mathcal{P}|\to 0} (z_n^2 - z_0^2) = \frac{1}{2} \lim_{|\mathcal{P}|\to 0} (Z^2 - z_0^2).
\end{aligned}$$

In particular, if L is a closed curve, so that $Z = z_0$, we again find that (10) vanishes.

Example 3. Part 1.

Example

Suppose $f(z) = 1/(z-a)$, and let L be the curve with equation

$$z = a + re^{it} \quad (0 \le t \le 2\pi),$$

i.e., the circle of radius r with center at the point a, traversed once in the positive direction. Dividing L into n equal arcs $\sigma_k = \widehat{z_{k-1}z_k}$, where

$$z_k = a + re^{2k\pi i/n} \quad (k = 0, 1, \ldots, n)$$

($z_n = z_0$, since L is closed), we choose ζ_k to be the midpoint of the arc σ_k:

$$\zeta_k = a + re^{(2k-1)\pi i/n} \quad (k = 1, \ldots, n).$$

As a result,

$$f(\zeta_k) = \frac{1}{\zeta_k - a} = \frac{1}{r} e^{-(2k-1)\pi i/n},$$

and

Example 3. Part 2.

Example

$$\sum_{k=1}^{n} f(\zeta_k)(z_k - z_{k-1}) = \sum_{k=1}^{n} e^{-(2k-2)\pi i/n}\left(e^{2k\pi i/n} - e^{(2k-1)\pi i/n}\right)$$
$$= \sum_{k=1}^{n}\left(e^{\pi i/n} - e^{-\pi i/n}\right)$$
$$= 2i\sum_{k=1}^{n}\sin\frac{\pi}{n} = 2i\, n\sin\frac{\pi}{n}.$$

It follows that

$$\oint_L \frac{dz}{z-a} = \lim_{n\to\infty} 2i\, n\sin\frac{\pi}{n}$$
$$= 2\pi i \lim_{n\to\infty}\frac{\sin\pi/n}{\pi/n} = 2\pi i.$$

Properties of complex integrals. Part 1.

Next we derive some elementary properties of complex integrals. The first three properties can be verified immediately either by using (12) to go over line integrals, or by using (11) directly. In each case, we assume that L is a rectifiable curve, and that $f(z)$ is defined and continuous on L.

- $\int_L f(z)dz = -\int_{-L} f(z)dz$,
  where L denotes the curve L traversed in the opposite direction.
- Suppose that

$$L = L_1 + L_2 + \cdots + L_m,$$

  where the final point of L_k coincides with the initial point of L_{k+1} $(k = 1, \ldots, m-1)$. Then we have

$$\int_L f(z)dz = \int_{L_1} f(z)dz + \int_{L_2} f(z)dz + \cdots + \int_{L_m} f(z)dz.$$

Properties of complex integrals. Part 2.

- $\int_L (c_1 f_1(z) + \cdots + c_m f_m(z))\, dz = c_1 \int_L f_1(z) dz + \cdots + c_m \int_L f_m(z) dz$, where $f_1(z), \cdots, f_m(z)$ are arbitrary complex functions which are defined and continuous on L, and $c_1, \ldots, c_m$ are arbitrary complex constants.
- Suppose $f(z)$ satisfies the inequality

$$|f(z)| \leq M$$

al all points of the curve L, e.g., suppose M is the quantity

$$\max_{z \in L} |f(z)|$$

Then we have the estimate

$$\left| \int_L f(z) dz \right| \leq Ml, \tag{13}$$

where l is the length of the curve L.

The case of smooth curves. Part 1.

Suppose L is smooth, i.e. suppose there is at least one parametric representation (1) such that $\lambda(t)$ has a continuous nonvanishing derivative $\lambda'(t)$ at every point of the interval $[a, b]$. Geometrically $\lambda'(t_0) \neq 0$ means that L has a tangent inclination $Arg\, \lambda'(t_0)$ at the point $z_0 = \lambda(t_0)$. Therefore a smooth curve has a tangent at every point, and moreover the inclination of the tangent varies continuously as the point of tangency moves along the curve. A smooth curve is rectifiable, with length

$$l = \int_a^b \sqrt{[\mu'(t)]^2 + [\nu'(t)]^2} dt \tag{14}$$

The case of smooth curves. Part 2.

Since

$$\lambda'(t) = \mu'(t) + i\nu'(t), \quad |\lambda'(t)|^2 = [\mu'(t)]^2 + [\nu'(t)]^2,$$

(14) can also be written as

$$I = \int_a^b |\lambda'(t)| dt.$$

Similarly, (8) becomes

$$\int_a^b \{P[\mu(t), \nu(t)]\mu'(t) + Q[\mu(t), \nu(t)]\nu'(t)\} \, dt \tag{15}$$

The case of smooth curves. Part 3.

while (12) takes the form

$$\begin{aligned}\int_L f(z)dz &= \int_a^b \{u[\mu(t),\nu(t)]\mu'(t) + v[\mu(t),\nu(t)]\nu'(t)\}\,dt + \\ &+ i\int_a^b \{v[\mu(t),\nu(t)]\mu'(t) + u[\mu(t),\nu(t)]\nu'(t)\}\,dt \\ &= \int_a^b \{u[\mu(t),\nu(t)] + iv[\mu(t),\nu(t)]\}\{\mu'(t) + i\nu'(t)\}\,dt \\ &= \int_a^b f[\lambda(t)]\lambda'(t)dt.\end{aligned} \tag{16}$$

The case of smooth curves. Example.

Example

Formula (16) can be used to simplify the evaluation of the integrals such as in the previous examples. In fact, since

$$z = \lambda(t) = a + r\,e^{it}, \quad \lambda'(t) = i\,r\,e^{it},$$

we have

$$\int_L \frac{dz}{z-a} = \int_0^{2\pi} \frac{ire^{it}}{re^{it}}\,dt = i\int_0^{2\pi} dt = 2\pi i.$$

The case of smooth curves. Part 4.

The formulas just written for the case of a smooth curve L remain valid if L is merely *piecewise smooth*, i.e., if $[a, b]$ can be divided into a finite number of subintervals

$$a = a_0 \leq t \leq a_1, a_1 \leq a_2, \ldots, a_{m-1} \leq t \leq a_m = b$$

such that $\lambda(t)$ has a continuous nonvanishing derivative on each subinterval, but not necessarily on $[a, b]$ itself. Thus $\lambda'(t)$ is now allowed to have discontinuities at finitely many interior points on $[a, b]$, provided nonzero one-sided derivatives (and hence one-sided tangents) exist at every such point.

Thank you